Multimodal Misinformation Detector Agent

Data Science and AI Lab Project

Milestone 4 Report

Group 9

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Contents

1. Objective

The objective of this milestone is to evaluate the trained models using appropriate performance metrics to assess their accuracy, reliability, and generalization capabilities.

Furthermore, the milestone aims to discuss the limitations of the current approach, including data constraints, model shortcomings, and evaluation challenges, while proposing possible improvements to enhance model robustness, interpretability, and overall effectiveness in future iterations.

2. Evaluation setup

2.1 Test Datasets

- Image data
  - **Source -** Kaggle – AI-Generated Images vs Real Images
  - **Structure:** Binary labelled dataset (*real* and *fake*) divided into train and test folders.
  - **Sample count:** 48,000 train, 12,000 test images.
- Text data
  - **Source:** HuggingFace - FEVER dataset (Fact Extraction and VERification)
  - **Structure:** Each entry contains a *claim*, *label*, and extracted *evidence sentences* from Wikipedia.
  - **Sample count:** 20,000 balanced samples after filtering and sampling equal counts for *SUPPORTS* and *REFUTES*.
- **Preprocessing:**
  - Removed instances labeled *NOT ENOUGH INFO*..
  - Filtered and chose claims with 1–10 evidence sentences.
  - Sampled 10,000 examples per label for balanced training.
  - Tokenization using microsoft/deberta-v3-base tokenizer (max length = 256).

2.2 Environment and Evaluation setup

- **Environment:** Kaggle Notebook and Google Colab with NVIDIA T4 GPU
- **Frameworks:** PyTorch, Hugging Face Transformers.
- **Loss Functions:** Cross-entropy (text), BCEWithLogitsLoss (image).
- **Training strategy:**
    - Gradient accumulation with early stopping.
    - Learning rate scheduling and weight decay regularization.
    - FP16 mixed-precision training for efficiency.
    - Lora adapter added to gemma3-270m to train it more efficiently with available GPU compute power.
- Training arguments of Gemma-3-270m model

```
args = TrainingArguments(
    output_dir=OUTPUT_DIR,
    learning_rate=2e-4,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    gradient_accumulation_steps=1,
    num_train_epochs=1,
    weight_decay=0.05,
    logging_steps=50,
    eval_strategy="epoch",
    save_strategy="epoch",
    save_total_limit=2,
    load_best_model_at_end=True,
    metric_for_best_model="f1",
    greater_is_better=True,
    bf16=(dtype == torch.bfloat16),
    fp16=(dtype == torch.float16),
    report_to="none",
)
```

3. Performance metrics

- **Metrics:**
  - Accuracy.
    - **Easy to interpret:** It provides a simple measure of how often the model's predictions are correct.
    - **Good for balanced datasets:** When all classes have roughly equal representation, accuracy gives a reliable overall performance measure.
    - **Fast to compute:** It's computationally inexpensive and widely supported in evaluation frameworks.
  - F1-Score
    - **Balances precision and recall:** It's useful when both false positives and false negatives are important to minimize.
    - **Effective for imbalanced data:** Provides a more realistic view of performance when one class dominates the dataset.
    - **Class-wise insight:** Can be computed per class (macro/micro F1) to reveal detailed model behavior.

- **Accuracy** gives a general sense of overall correctness — useful if your dataset is balanced.

  F1 Score gives a fairer picture for **imbalanced** or **class-sensitive tasks**, ensuring both false positives and false negatives are considered.

  By combining both metrics, the evaluation offers a comprehensive understanding of each model's strengths and weaknesses in terms of overall accuracy and balanced predictive capability.

4. Limitations

- **Computational Constraints:** Gemma 3–270M could not be fully trained due to GPU memory/time limitations.
- **Generalization:** Models were tested on static datasets; real-world inputs might vary in style, tone, or noise. This can be corrected by augmenting the existing dataset.

5. Proposed Improvements / Next Steps

- **Data Augmentation:** Introduce paraphrasing for text and transformations for images to improve robustness.
- **Extended Training:** Re-train Gemma-3-270m for more epoch.
-

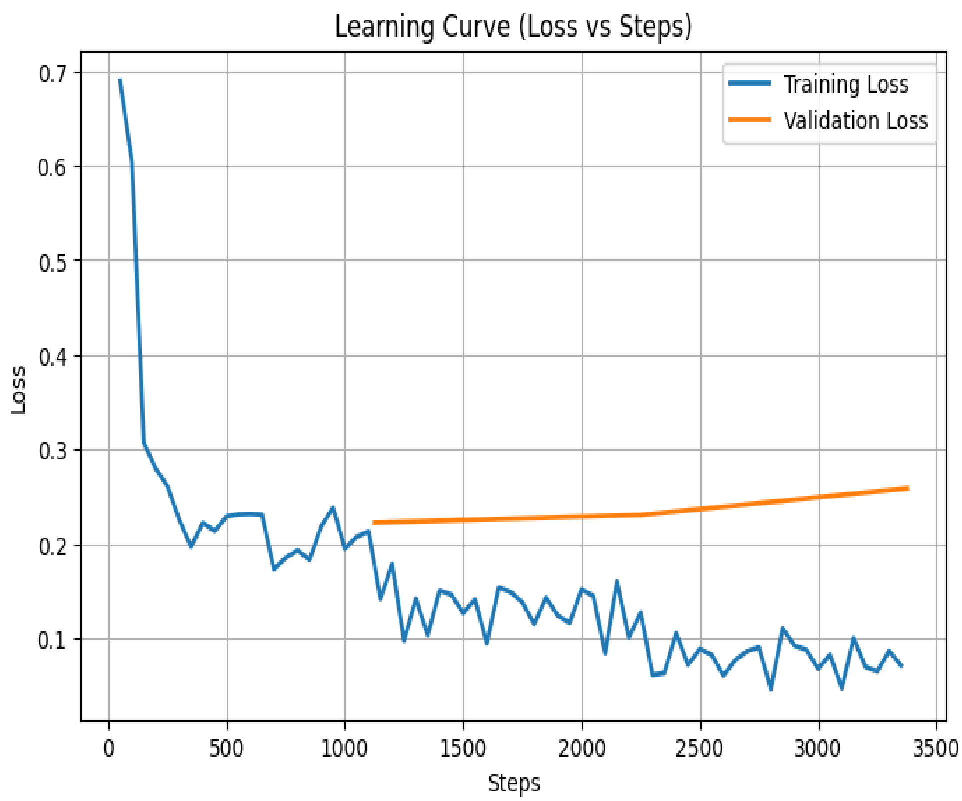
6. Quantitative Results

7.1 Text Models

- Metrics of DeBERTa model

Epoch	Training Loss	Validation Loss	Accuracy	F1
1	0.213000	0.222147	0.948000	0.948361
2	0.127300	0.230390	0.950000	0.950199
3	0.071100	0.258387	0.950000	0.950446

```
📊 Final Evaluation Metrics:
eval_loss : 0.2584
eval_accuracy: 0.9500
eval_f1    : 0.9504
eval_runtime: 13.8990
eval_samples_per_second: 143.8950
eval_steps_per_second: 4.5330
epoch      : 3.0000
```



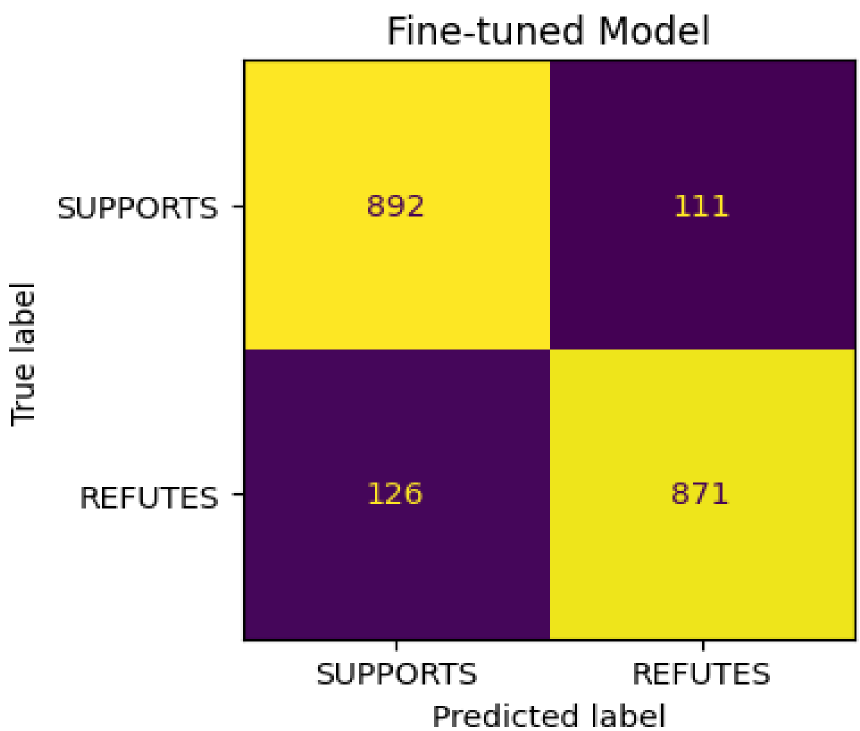
- Metrics of Gemma3-270m model Finetuned on the fever dataset.

[1125/1125 40:29, Epoch 1/1]

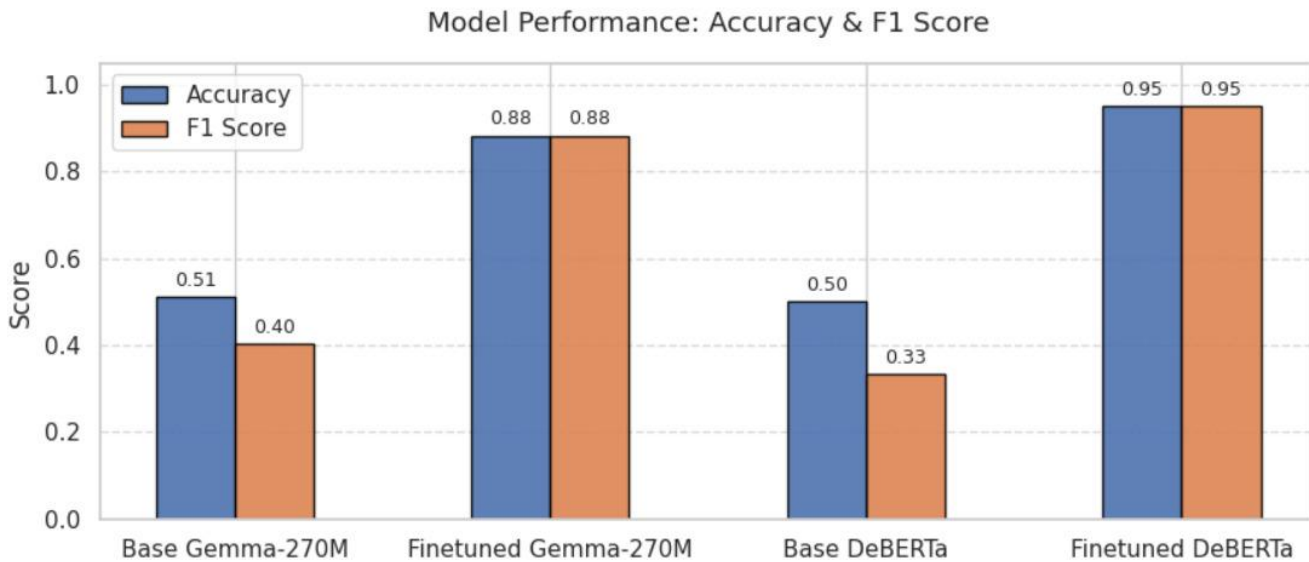
Epoch	Training Loss	Validation Loss	Accuracy	F1	Precision	Recall
1	0.291800	0.303256	0.881500	0.880243	0.886965	0.873621

```
... {'eval_loss': 0.303255558013916,
     'eval_accuracy': 0.8815,
     'eval_f1': 0.8802425467407782,
     'eval_precision': 0.8869653767820774,
     'eval_recall': 0.8736208625877633,
     'eval_runtime': 101.2831,
     'eval_samples_per_second': 19.747,
     'eval_steps_per_second': 1.234,
     'epoch': 1.0}
```

```
=== Model After Fine-tuning ===

              precision    recall  f1-score   support

    SUPPORTS     0.8762    0.8893    0.8827      1003
     REFUTES     0.8870    0.8736    0.8802       997

    accuracy                         0.8815      2000
   macro avg     0.8816    0.8815    0.8815      2000
weighted avg     0.8816    0.8815    0.8815      2000
```



- Comparison of DeBERTa and Gemma3-270m models.



7.2 Image Models

- Clip-vit-base-patch32

```
Epoch 3 Summary:
  Train Loss: 0.1154
  Validation Loss: 0.1089
  Validation Accuracy: 0.9591
🎉 New best model saved! Accuracy: 0.9591 at Epoch 3
   Model saved to /kaggle/working/best_clip_finetuned_classifier.pth

Training complete.
The best model with accuracy 0.9591 is saved at /kaggle/working/best_clip_finetuned_classifie
r.pth
```

  - Achieved ~0.95 validation accuracy after 3 epochs.
  - Observed clear separation between real and AI-generated images.
  - Model stable with dropout regularization, avoiding overfitting despite limited dataset size.

- Convnextv2-base-22k-224
  - Achieved ~0.92 validation accuracy after 3 epochs.

7. Artifacts and reproducibility

- **DeBERTA Model Checkpoints:** *fever_finetuned_deberta/*
- **Clip Model Checkpoints:** *clip_finetuned_classifier.pth*
- **Gemma3-270m Model**: *Gemma_archive.zip*
- **Tokenizer and Processor:** Hugging Face pre-trained modules.
- **Training Logs:** Generated via Hugging Face Trainer logs and PyTorch loss curves.
- **Reproducibility:** Training code pushed into GitHub repository.

8. Summary

This milestone demonstrated that DeBERTa and CLIP both achieved strong generalization on unseen data using accuracy and F1 score metrics.
While DeBERTa excelled in capturing textual semantics and CLIP showed robustness in visual classification, Gemma 3–270M remains promising but was limited by compute resources.
Future improvements will focus on extended fine-tuning, data augmentation, and interpretability techniques to further enhance performance and deployment readiness.